Physical Device Test Log & Verification Audit

Standalone Android APK Execution, Offline Resilience & Persistence Testing

Student Name: Mahgoub Adil Ahmed Alhassan
Registration Number: 25/28013
Verification Code: MOB-A2-8013
Application Name: MahgoubPortfolio
Package Identifier: rw.ac.ines.ug2528013.mahgoubportfolio
Build Pipeline: Expo EAS Build (Preview APK profile)
Target Test Device: Samsung Galaxy A53 5G (Physical Hardware)
Android OS Version: Android 14 (One UI 6.0, API Level 34)
Screen Dimensions: 1080 x 2400 px (405 ppi, 20:9 Aspect Ratio)
Audit Timestamp: 12 October 2026, 12:45 CAT

1. Comprehensive Test Execution Matrix

Test ID	Feature / Scenario	Test Procedure	Expected vs. Actual Outcome	Status
TC-01	**Standalone APK Installation**	Download `MOB_A2_2528013.apk` directly via USB and install via Android Package Installer without Expo Go.	App installed cleanly with custom app icon and package name `rw.ac.ines.ug2528013.mahgoubportfolio`. Launched independently.	**PASS**
TC-02	**Identity Statement Accuracy**	Inspect line 1 of Home Screen upon launch.	Displayed exact mandatory sentence: *"My name is Mahgoub Adil Ahmed Alhassan."* Verification code `MOB-A2-8013` visible.	**PASS**
TC-03	**Airplane Mode (Offline Readiness)**	Enable Airplane Mode on physical Android device (disable Wi-Fi and Cellular). Launch app and navigate tabs.	Home, Skills, Projects, Profile Editor, and About screens render instantaneously with zero network errors. Device clock confirmed.	**PASS**
TC-04	**External Link Labeling & Safety**	Tap GitHub repository link while in Airplane Mode.	Triggered confirmation modal warning that internet connectivity is required. Did not crash or hang.	**PASS**
TC-05	**Project Stack Navigation**	Navigate from Projects List to *INES Attendance Tracker* detail screen, then press Android hardware Back button.	Project detail opened with deep architectural breakdown. Back button popped screen cleanly while preserving active bottom tab state.	**PASS**
TC-06	**Validation Rule 1 (Headline)**	In Profile Editor, input "Dev" (3 chars) into Headline and attempt to save.	Save blocked. Displayed inline error: "*Headline must be at least 6 characters long (currently 3).*"	**PASS**
TC-07	**Validation Rule 2 (Biography)**	In Profile Editor, input "Student." (8 chars) into Biography and attempt to save.	Save blocked. Displayed inline error: "*Biography is too short. Minimum 20 characters required.*"	**PASS**
TC-08	**Valid Input & AsyncStorage Save**	Input valid headline (42 chars) and bio (120 chars). Tap "Save Profile Changes".	Validation passed. Success toast rendered. AsyncStorage updated with serialized JSON payload.	**PASS**
TC-09	**Process Kill & Cold Restart Persistence**	Swipe away app from Android Recent Apps task manager to terminate process. Relaunch app.	Home Screen and Profile Editor loaded custom headline and biography from AsyncStorage without data loss.	**PASS**
TC-10	**Media Picker Permissions & Cancellation**	Tap "Gallery Selection", then cancel picker without selecting an image.	Cancellation handled gracefully without unhandled exceptions. Selected local image URI persisted and rendered in circular avatar.	**PASS**

		Then select image on second attempt.		
TC-11	**Large-Text Scalability (130% Font)**	Set Android System Font size to "Largest" in OS Accessibility Settings.	Flexbox cards expanded dynamically. No overlapping text or clipped verification badges detected.	**PASS**

2. Hardware Test Sign-off & Audit Conclusion

The standalone Android APK for **MahgoubPortfolio** satisfies 100% of the SWE 3409 physical device execution requirements. The application demonstrates resilient offline execution, verifiable data persistence across cold restarts, and strict controlled input validation.

Lead Student Engineer: Mahgoub Adil Ahmed Alhassan | **Date:** 12 October 2026 | **Verification Code:** MOB-A2-8013